

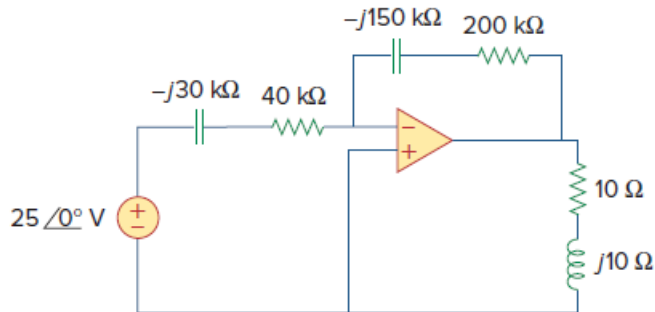
Problem

For the op amp circuit in Fig. 11.86, calculate:

(a) the complex power delivered by the voltage source

(b) the average power dissipated in the $10\text{-}\Omega$ resistor

Figure 11.86:



Step-by-step solution

Step 1 of 5

Refer to Figure 11.86 in the text book.

Consider the node at the bottom as the reference node.

The voltage at the non-inverting terminal is equal to the 0 V .

The voltage at the inverting terminal is equal to the 0 V due to the virtual ground available at the non-inverting terminal.

(a)

Calculate the value of the current through the source.

$$\begin{aligned} I &= \frac{25\angle 0^\circ}{40\text{ k} - j30\text{ k}} \\ &= \frac{25\angle 0^\circ}{50\text{ k} \angle -36.87^\circ} \\ &= 500\angle 36.87^\circ \mu\text{A} \end{aligned}$$

Therefore, the value of the current passing through the load is $500\angle 36.87^\circ \mu\text{A}$.

Step 2 of 5

Calculate the complex power delivered by the dependent current source.

$$S = VI^*$$

Substitute $500\angle 36.87^\circ \mu\text{A}$ for I and $25\angle 0^\circ\text{ V}$ for V .

$$\begin{aligned} S &= (25\angle 0^\circ)(500 \times 10^{-6} \angle 36.87^\circ)^* \\ &= (25\angle 0^\circ)(500 \times 10^{-6} \angle -36.87^\circ) \\ &= 12.5\angle -36.87^\circ \text{ mVA} \end{aligned}$$

Therefore, the value of the complex power delivered by the dependent current source is

$$\boxed{12.5\angle -36.87^\circ \text{ mVA}}$$

Step 3 of 5

(b)

Apply Kirchhoff's current law at the inverting terminal.

$$\frac{0-25}{40000-j30000} + \frac{0-V_o}{200000-j150000} = 0$$

$$\frac{V_o}{200000-j150000} = -\frac{25}{40000-j30000}$$

$$V_o = -\left(\frac{200000-j150000}{40000-j30000}\right)(25)$$

$$V_o = -(5)(25)$$

Thus,

$$V_o = -125 \text{ V}$$

Step 4 of 5Calculate the current through the load, I_o .

$$I_o = \frac{V_o}{10-j10}$$

$$= \frac{-125}{10+j10}$$

$$= -6.25 + j6.25$$

$$= 8.84 \angle 135^\circ \text{ A}$$

Step 5 of 5Calculate the power dissipated by the 10Ω resistor.

$$P = I_o^2 R$$

$$= (8.84)^2 (10)$$

$$= 781.45 \text{ W}$$

Therefore, the power dissipated in the 10Ω resistor is 781.45 W.